CMPUT 466 Course Project

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1 Introduction

The goal of this project is predicting the weather in Edmonton solely from the date and determining if this is a reasonable task. The dataset was obtained from the City of Edmonton's Open Data Portal, specifically the Weather Data (Daily) package from Environment Canada. This dataset was then stripped to only contain entries from Edmonton. Removing all entries not associated with the Edmonton International Airport Station resulted in 4559 rows of data, spanning from 2000 to 2024.

2 Formulation

The inputs are the year, month, day. The output is the mean temperature, in °C. I had originally planned to include other factors such as amount of precipitation and wind speed in the inputs which would very likely increase the accuracy significantly, but was curious to see how well the temperature could be predicted from the date alone.

3 Approaches

3.1 Algorithms

The 3 machine learning algorithms chosen for this task were:

1. Stochastic Gradient Descent Regressor
2. Random Forest Regressor
3. Multi-layer Perceptron Regressor

Each algorithm used is from sklearn's libraries. The hyperparameters for each algorithm and how they were tuned are:

1. α, tuned by selecting the lowest root mean squared error from each alpha from $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$
2. n_estimators, tuned by selecting the lowest root mean squared error from each n_estimators from $\{100, 200, 500, 800\}$
3. α, tuned by selecting the lowest root mean squared error from each alpha from $\{0.00001, 0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1\}$

3.2 Training

The original data is modified to drop all columns except year, month, day, and mean temperature. The predictors (year, month, and day) are then scaled using the sklearn StandardScaler. This data is then split into training and testing datasets using the train_test_split sklearn function, where 20% of the data is designated for testing.

4 Evaluation

4.1 Success

To measure the success of the model, I used root mean square error (RMSE). The loss function used was the default for the algorithms used, squared error. I chose to use RMSE over mean absolute error to measure the suceess of the model as RMSE is

more affected by large differences, which would be very important to penalize in our situation; predicting the mean temperature and being off by a significant amount is not acceptable.

4.2 Goal

The goal of this project was to predict the mean temperature of a day, given only the day. Unfortunately, I believe this to only be an approximation as it is not feasible to predict the temperature only given the day and no other details (such as wind speed, precipitation, and cloud cover). However, I believe this approximation to be accurate as my measure of success returned a reasonable value, where being off by only a couple of units wouldn't really matter much for something such as the temperature.

5 Results

5.1 Baseline

For the baseline, I computed the mean of all temperatures, and that became the prediction for all inputs. This unsurprisingly performed very poorly, which is to be expected as it only exists to function as a baseline. The RMSE was 12.2504.

5.2 Stochastic Gradient Descent

This was the first algorithm used and unsurprisingly it performed quite poorly, almost on par with the baseline. Fitting the data to a line is a poor choice as the weather throughout the year is not linear at all. The lowest RMSE achieved was 12.0887, with an α of 0.03. This value is unacceptable, as predicting the weather and being wrong by 12 °C is completely unreasonable.

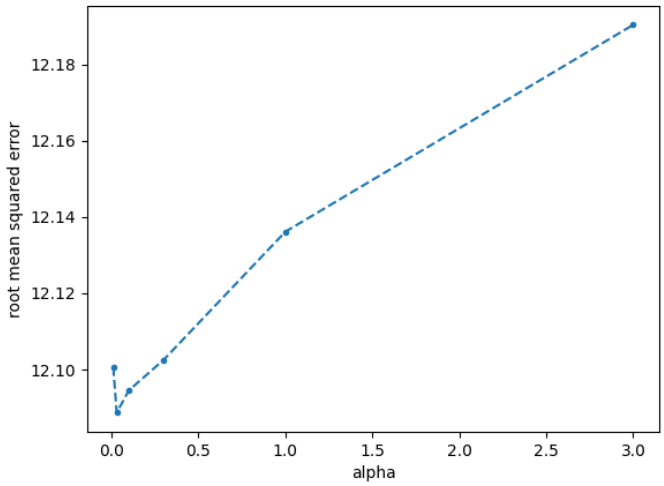


Figure 1: RMSE for each α tested from $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$

5.3 Random Forest Regression

This is the second algorithm used and it performed well. This was a massive improvement from the previous SGD algorithm, providing much more accurate results. The hyperparameter tuned was n_estimators, which controls the amount of trees used. The RMSE was not impacted much by tuning n_estimators, but the lowest value was achieved with a n_estimators of 200. Higher values took longer to compute and performed slightly worse. I was very pleased with this performance as predicting the weather and only being off by $\sim$4 °C (especially considering large mistakes are penalized more heavily by RMSE) is quite good, considering the lack of contextual information provided for the prediction.

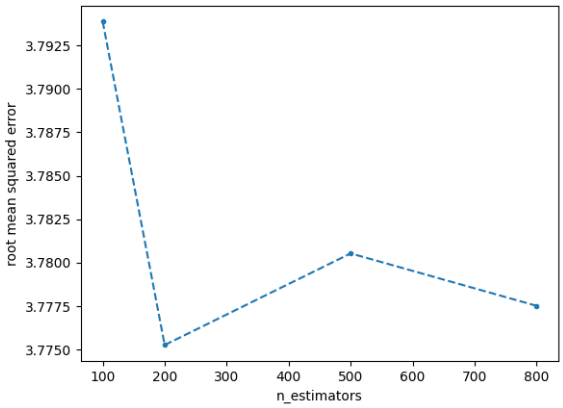


Figure 2: RMSE for each n_estimators tested from $\{100, 200, 500, 800\}$

5.4 MLP Regressor

The last algorithm used was the Multi-layer Perceptron regressor from sklearn. This model performed better than SGD, but worse than Random Forest Regression. The hyperparameter tuned was alpha, which ultimately didn't affect the RMSE all that much. The best RMSE returned was 5.897214 with an α of 0.00001, with a second best RMSE of 5.897215 with an α of 0.0001. With an RMSE of $\sim$6 °C, this model borders on providing an unacceptable estimate. Again, considering the fact that additional context isn't provided for predicting the temperature, I would consider being 6 °C off just from the date alone not a bad estimate.

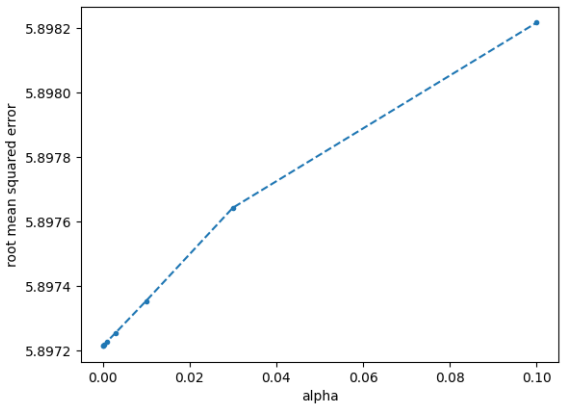


Figure 3: RMSE for each α tested from $\{0.00001, 0.0001, 0.0003, 0.001, 0.003, 0.01, 0.03, 0.1\}$

6 Interpretation of Results

Looking at the performance of each algorithm, the results were not very surprising. The baseline and SGD algorithms performed extremely poorly as the formulation of these do not fit the task at hand. The performance of Random Forest Regression was surprising, as it performed quite well given the task of predicting the weather from only the date without any additional context. The MLP Regressor performed okay, but is not the algorithm I would use for this specific task.

Considering the somewhat unrealistic task and the sporadic nature of the weather in general, I am quite pleased by the results I achieved. My predictions would have obviously been more accurate if I had also included other factors in my predictions, but I found this to be more interesting.